

K0WLY Two-Way CW Keyer

LilyGO T-Display S3 AMOLED

Operator's Manual — Preliminary

Version 1.0

Call Sign: K0WLY

Grid Square: DN40 — Saratoga Springs, Utah

License and Attribution

Copyright © 2026 K0WLY (Carl Cowley)

Saratoga Springs, Utah — Grid Square DN40

Amateur Radio Callsign: K0WLY (Extra Class)

CERN Open Hardware Licence Version 2

Weakly Reciprocal (CERN-OHL-W v2)

This work — including all firmware source code, schematic diagrams, PCB design files, and documentation — is licensed under the CERN Open Hardware Licence Version 2 – Weakly Reciprocal (CERN-OHL-W v2).

You may redistribute and modify this work and make products using it under the terms of the CERN-OHL-W v2 (https://ohwr.org/cern_ohl_w_v2.txt).

Documentation is additionally licensed under Creative Commons Attribution 4.0 International (CC BY 4.0).

Attribution Requirements:

- The copyright notice above must be retained on all copies and derivatives
- Attribution to K0WLY must appear prominently in any derivative work
- Any modified versions must be clearly marked as such
- The original project source must be acknowledged in derivative documentation

73 de K0WLY

1. Overview

The K0WLY Two-Way CW Keyer is an ESP32-S3 based iambic keyer built on the LilyGO T-Display S3 AMOLED board featuring a 1.91" 536×240 AMOLED display. It provides full iambic keying with sidetone, Morse code decoding displayed in real time, and peer-to-peer two-way CW communication over WiFi using ESP-NOW — no router or network infrastructure required.

Key features:

- Iambic Mode A keyer with adjustable speed (5–40 WPM)
- Straight key mode selectable via hardware switch
- Adjustable sidetone frequency (400–900 Hz) — each unit independent
- Logarithmic volume control for speaker and headphones
- Head copy delay (0–3 seconds) for incoming characters
- Automatic peer-to-peer WiFi pairing via ESP-NOW
- Two-way CW: transmit and receive simultaneously
- Full Morse code decoding: letters, numbers, punctuation, prosigns
- Frame buffer display — instantaneous updates, no glitch
- Non-volatile settings storage (survives power cycles)
- Single pot with mode cycling for all parameter adjustments

2. Display Layout

The AMOLED display is divided into four zones from top to bottom:

Zone	Location	Content
------	----------	---------

Header Bar	Top 32px	Callsign, WPM, Freq, Delay, Vol, SK/IAM, SOLO/DUAL
TX Line	Upper middle	Outgoing decoded characters (green, scrolling left)
RX Line	Lower middle	Incoming decoded characters (cyan, scrolling left)
Status Area	Bottom 96px	Active pot mode, current value, peer MAC address

2.1 Header Bar

The header bar displays all key operating parameters simultaneously:

- K0WLY — Your callsign (left side, green)
- WPM — Current speed, highlighted yellow when selected
- Hz — Sidetone frequency, highlighted cyan when selected
- DLY — Head copy delay in seconds, highlighted orange when selected
- VOL — Volume percentage, highlighted magenta when selected
- IAM / SK — Current key mode (Iambic or Straight Key)
- SOLO / DUAL — WiFi peer status (white=solo, green=dual)

When a parameter is selected but not in edit mode it appears in a dim version of its color. When in edit mode it brightens to full color and [EDIT] appears in the status area.

2.2 TX and RX Lines

Decoded characters scroll from right to left on each line. The TX line (green) shows characters you are sending. The RX line (cyan) shows characters received from the peer unit, subject to any head copy delay you have set.

3. Controls and Connections

3.1 GPIO Pin Assignments

Function	GPIO	Notes
DIT Paddle	GPIO11	Active LOW, ext 10kΩ pull-up to 3.3V
DAH Paddle	GPIO10	Active LOW, ext 10kΩ pull-up to 3.3V

Key Out	GPIO12	HIGH = keyed, drives PC817 optocoupler
Sidetone	GPIO13	PWM → 470Ω → 2N4401 → Speaker/Headphones
Pot (ADC)	GPIO14	Wiper of 10–20kΩ pot
Key Mode	GPIO15	SPST switch: open = iambic, closed = straight key
Paddle Reverse	GPIO16	PBNO momentary button
Pot Mode Select	GPIO39	PBNO momentary button

3.2 Paddle Input

The keyer accepts a standard two-lever iambic paddle. Connect the dit lever to GPIO11 and the dah lever to GPIO10 with external 10kΩ pull-up resistors to 3.3V on each line. The paddle contacts should close to ground.

NOTE: *If your paddle levers feel reversed, press the Paddle Reverse button (GPIO16) to swap them in software without rewiring.*

3.3 Key Output

GPIO12 drives a PC817 optocoupler to key your transceiver. The optocoupler provides full electrical isolation between the ESP32-S3 and the radio's key line, which may carry up to 9V on some transceivers including the Kenwood TS-120S.

3.4 Sidetone Output

The sidetone is a PWM square wave on GPIO13 driving a 2N4401 NPN transistor. A bi-polar electrolytic coupling capacitor is used on the headphone output only.

Speaker circuit — direct connection, NO coupling capacitor:

3.3V → Speaker → Collector / 2N4401 #1 / GPIO13 → 470Ω → Base / Emitter → GND

NOTE: *Do NOT put a coupling capacitor in series with the speaker. It will completely block all audio. The speaker connects directly between 3.3V and the transistor collector.*

Headphone circuit — separate transistor with coupling cap:

3.3V → 1kΩ → Collector / 2N4401 #2 (base shared) / Emitter → GND. Collector → 47μF bi-polar cap → 100Ω → 3.5mm tip. 3.5mm sleeve → GND.

3.5 Key Mode Switch (GPIO15)

A SPST latching switch. Open (pulled high) = iambic mode. Closed (GPIO15 to GND) = straight key mode, where the dit paddle acts as a simple key-down/key-up and the dah paddle is ignored.

3.6 Paddle Reverse Button (GPIO16)

A PBNO momentary pushbutton. Each press toggles the dit/dah assignment in software. Resets to normal on power cycle.

3.7 Pot Mode Select Button (GPIO39)

A PBNO momentary pushbutton. Short press cycles parameter modes. Long press enters/exits edit mode. See Section 4.

4. Parameter Adjustment

All four adjustable parameters are controlled by a single potentiometer. The Pot Mode Select button cycles through parameters and activates edit mode.

4.1 Button Press Functions

Press Type	Action	Result
Short press	Cycle mode	WPM → FREQ → DELAY → VOL → WPM
Short press (in edit)	Exit edit	Saves value, exits edit mode
Long press (1 sec)	Toggle edit	Enters or exits edit mode

4.2 Edit Mode

The pot only affects values when in edit mode, preventing accidental changes.

- Short press to select the desired parameter
- Long press (1 second) to enter edit mode — [EDIT] appears in status area
- Turn pot to adjust — saves to flash in real time
- Short or long press to exit edit mode

NOTE: All settings save automatically to flash as you adjust. Values restore on next power cycle.

4.3 Parameter Ranges

Parameter	Range	Notes
WPM	5 – 40 WPM	PARIS standard (1200/WPM ms per dit)
Frequency	400 – 900 Hz	Sidetone pitch, each unit independent
Head Copy Delay	0.0 – 3.0 sec	Incoming char display delay, 0.1s steps
Volume	0 – 100%	Logarithmic curve, max PWM duty 78%

5. WiFi Peer-to-Peer Operation

5.1 Auto-Discovery

The keyer uses ESP-NOW for direct peer-to-peer communication without a WiFi router. On power-up each unit broadcasts a discovery packet every 2 seconds. When a peer responds the header shows DUAL in green and the peer MAC appears in the status area.

5.2 Transmitting to Peer

Every element keyed (dit or dah) transmits to the peer unit in real time. The peer plays back each element at its own locally set sidetone frequency. Each operator always hears code at their own preferred pitch, completely independent of the other station's setting. Decoded characters also transmit and appear on the peer's RX line.

5.3 Independent Sidetone Frequency

Each unit plays all CW — both sent and received — at its own locally configured sidetone frequency. If one unit is set to 550 Hz and the other to 700 Hz, each operator hears everything at their own pitch. This is exactly how real CW operation works — each operator sets their sidetone to their personal preference, independent of the other station.

5.4 Head Copy Delay

The head copy delay applies only to the display of incoming decoded characters. Audio playback of incoming elements is always immediate. This lets you practice mentally decoding received code before the character appears on screen.

NOTE: Set delay to 0 for normal operation. Increase gradually as your head copy skill improves.

6. Operating Tips

6.1 Getting Started

- Power on both units and wait approximately 5 seconds for WiFi to initialize
- Verify header shows desired WPM before keying
- SOLO = no peer found yet, DUAL = paired and ready
- TX line (green) shows your outgoing characters
- RX line (cyan) shows incoming characters from peer

6.2 Iambic Keying

- Brief press — sends one element (E or T)
- Hold dit paddle: E → I → S → H → 5
- Hold dah paddle: T → M → O → 0
- Squeeze both paddles — alternates dit-dah for A, C, K, R, etc.

6.3 CW Timing Reference

Element	Duration	At 20 WPM (60ms dit)
Dit	1 × dit length	60 ms
Dah	3 × dit length	180 ms
Inter-element gap	1 × dit length	60 ms
Character gap	3 × dit length	180 ms
Word gap	7 × dit length	420 ms

7. Factory Defaults

Parameter	Default Value
Speed	20 WPM
Sidetone Frequency	700 Hz
Head Copy Delay	0.0 seconds (off)

Volume	50%
Key Mode	Iambic (hardware switch)
Paddle Reverse	Normal (unreversed)
WiFi Channel	Channel 1

8. Firmware Information

Item	Details
Platform	ESP32-S3 (LilyGO T-Display S3 AMOLED)
Framework	Arduino via PlatformIO
Display Driver	RM67162 AMOLED, 536×240, QSPI
Display Library	LilyGo-AMOLED-Series v1.2.4
WiFi Protocol	ESP-NOW (peer-to-peer, no router)
Keyer Task	FreeRTOS task pinned to Core 1
WiFi / Display	Core 0 (Arduino loop)
Timer Resolution	1ms (vTaskDelayUntil)
Display Updates	Frame buffer in PSRAM, region flush
Settings Storage	ESP32 NVS (Preferences library)
CW Standard	PARIS (ITU-R M.1677-1)
License	CERN-OHL-W v2 / CC BY 4.0

9. Troubleshooting

Symptom	Likely Cause	Fix
No display on boot	Display init failed	Check USB, try BOOT button to reflash
Always SOLO	Peer out of range	Keep units within WiFi range, both on channel 1
Wrong characters	Paddles reversed	Press Paddle Reverse button (GPIO16)
No sidetone	Volume at minimum	Enter VOL edit mode and increase
Settings reset	NVS issue	Exit edit mode after adjusting
No sound to speaker	Coupling cap in circuit	Remove cap — speaker is direct connection

No key output	PC817 circuit issue	Check optocoupler wiring and GPIO12
---------------	---------------------	-------------------------------------

10. Morse Code Reference

The keyer decodes letters, numbers, punctuation, and prosigns. Prosigns: + = AR (end of message), = = BT (break), ~ = SK (end of contact).

Letters:

A · -	B - · · ·	C - · - ·	D - · ·	E ·	F · - · ·
G - - ·	H · · · ·	I · ·	J · - - -	K - · -	L · - · ·
M - -	N - ·	O - - -	P · - - ·	Q - - · -	R · - ·
S · · ·	T -	U · - -	V · · · -	W · - -	X - · · -
Y - · - -	Z - - · ·				

Numbers:

0 - - - - -	1 · - - - -	2 · · - - -	3 · · · - -	4 · · · · -	5 · · · · ·
6 - · · · ·	7 - - · · ·	8 - - - · ·	9 - - - - ·		

Punctuation:

· period · - - - -	, comma - - - - -	? question · - - - -	/ slash - - - - ·	(open - · - - ·	) close - - - - -
: colon - - - · ·	; semi - · - · · ·	- hyphen - · · · · -	' apos · - - - - ·	" quote · - · · · ·	@ at · - - · · ·
& amp · - · · ·	! excl - · - · -	_ under · - - - -			

Prosigns:

+ AR end msg · - · - ·	= BT break - · · · -	~ SK end QSO · · - · -	& AS wait · - · · ·		
------------------------	----------------------	------------------------	---------------------	--	--